

Mevlut Yildirim

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EDUCATION

Technical University of Munich

M.Sc. in Communications and Electronics Engineering

Dec. 2025

Munich, Germany

- **GPA: 1.2** (German scale: 1.0 best)
- **Relevant Coursework:** IoT Security, Network Security, Embedded Systems & Security, Smart Card Laboratory, IoT Remote Lab, Software Architecture Design for Distributed Embedded Systems
- **Thesis:** Physical Adversarial Attacks Using Fan-Based Holographic Projections

Hack The Box Academy

Cybersecurity Trainee

May 2025 – Present

- Completed Hack The Box job-role paths: Penetration Tester, SOC Analyst, and Junior Cybersecurity Analyst, with hands-on labs covering the full assessment workflow from reconnaissance to reporting.
- Focus areas proved by completed paths and modules: web exploitation (auth/session testing, input validation, XSS/SQLi, IDOR), Active Directory enumeration and attack paths, Linux/Windows privilege escalation, password/authentication attacks, network enumeration and traffic analysis.
- **Outcomes:** produced step-by-step write-ups, PoCs, and reusable playbooks; standardized checklists and report snippets aligned with OWASP WSTG/Top 10 and ATT&CK; practiced detection-aware operations and basic OPSEC in lab environments.
- **Tooling practiced:** Kali Linux, Burp Suite, OWASP ZAP, Nmap, Metasploit, Wireshark, ffuf, sqlmap, John the Ripper, Hashcat, Nessus, BloodHound, Impacket; automation with Python and Bash.

Istanbul Technical University

B.Sc. in Electronics and Communications Engineering

Jul. 2023

Istanbul, Turkey

- **GPA: 3.4/4.0**
- **Thesis:** ESP8266 Ad Hoc Mesh for Real-Time Swarm Coordination

WORK EXPERIENCE

German Aerospace Center (DLR)

Cybersecurity Research Intern

Aug. 2024 – Nov. 2024

Munich, Germany

- Conducted an offensive protocol assessment of SpaceWire, enumerated trust boundaries, and identified missing authentication/encryption and protocol weaknesses; outlined paths and mitigations.
- Built an adversary-centric threat model and attack-surface map for on-board networks; profiled attacker capabilities and preconditions to prioritize likely exploitation chains.
- Implemented adversary emulation in OMNeT++ to replay, spoof, inject, and disrupt traffic; captured evidence to demonstrate impact on integrity, availability and command authenticity.

Saportif Technology Inc.

IoT Systems Engineer / Sep. 2022 – Sep. 2023

Oct. 2020 – Sep. 2023

Istanbul, Turkey

- Developed an end-to-end IoT system with ESP32 and Raspberry Pi for a smart door-lock in shared offices, integrating sensors, actuators, RFID access control, and cloud-backed web interfaces.
- Secured the integration layer with HTTPS/TLS and authenticated REST APIs; mapped RFID identities to role-based access control and enforced server-side authorization for bookings and device commands.
- Implemented input validation, parameterized requests, rate limiting, and audit logging; documented issues using OWASP Top 10 categories to prioritize remediation.

Full-Stack Developer / Oct. 2020 – Sep. 2022

Istanbul, Turkey

- Built and maintained React and Node services with secure-by-default patterns and peer code reviews.
- Implemented authentication and session controls (RBAC, JWT, secure cookies), CSRF protection, and XSS mitigations including output encoding and CSP.
- Hardened REST APIs with input validation, parameterized queries, security headers, and structured error handling; controls aligned with OWASP Top 10 and WSTG.

Arcelik Inc.

Jul. 2022 – Sep. 2022

Software Engineer Intern

Istanbul, Turkey

- Built a Python test library that created a software-based virtual testing environment and removed reliance on hardware simulators.
- Achieved approximately 80% faster test execution on refrigerator control boards with reusable, automated test algorithms.

PROJECTS

Physical Adversarial Attacks Using Fan-Based Holographic Projections

May 2025 – Nov. 2025

- Planned and executed black-box security testing of traffic sign computer-vision pipelines; produced a threat model and attack-surface analysis for safety-critical scenarios.
- Built a programmable hardware proof of concept with a holographic POV fan and a repeatable test harness to run real-time experiments and collect evidence for vulnerability assessment.
- Quantified evasion/misclassification across lighting, distance, and angle, achieving up to 90% misclassification; delivered steps to reproduce, PoCs, and prioritized mitigations, and documented residual risk.

ESP8266 Ad Hoc Mesh for Real-Time Swarm Coordination

Oct. 2022 – Jul. 2023

- Built a decentralized ad hoc mesh using ESP8266 & ESP-NOW, removing the ground-station dependency via peer-to-peer links.
- Designed and implemented a real-time swarm coordination algorithm for dynamic environments and multi-node operation.
- Evaluated communication reliability and latency under varying node counts and mobility; documented results and limitations.

CERTIFICATIONS

- **Hack The Box Certified Junior Cybersecurity Analyst (CJCA)** **2026**

HONORS & AWARDS

- **IOTSEC CTF I – 1st place, TUM** **2024**
- **IOTSEC CTF II – 2nd place, TUM** **2024**
- **Kaspersky CTF – 4th place (Europe)** **2025**

SKILLS, TECHNOLOGIES & LANGUAGES

- **Technologies:** Python; C/C++; Bash; PowerShell; Linux/Kali Linux; Windows; Nmap; Burp Suite; Wireshark; OWASP ZAP; Metasploit; sqlmap; Nessus; BloodHound; Windows Active Directory.
- **Skills:** Penetration Testing; Vulnerability Assessment; Active Directory Enumeration, Secure Design Principles; Threat Analysis and Risk Assessment (TARA); Applied Cryptography; Web Application Security.
- **Languages:** Turkish (mother tongue); English: C1 (Advanced); German: B1 (Intermediate)